



Software Testing — Manual & Automation

Build a career in QA: manual testing foundations, Selenium automation, core Java, SQL, and the web technologies testers rely on daily.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

100 Days

Classroom + Live Training

14

Curriculum Modules

6

Hands-On Projects

50,000+

Testing Jobs Tracked on
LinkedIn

01 Learning Path

1

Manual Testing Fundamentals

Phase 1

SDLC models, white-box & black-box testing, defect life cycle, Agile (Scrum/Sprint) methodology, test plans, test cases, and STLC.

2

Automation Testing with Selenium

Phase 2

Selenium WebDriver architecture, locators, XPath & CSS selectors, handling web elements, waits, and the TestNG framework.

3

Core Java for Test Automation

Phase 3

Java basics, OOP, inheritance, exception handling, interfaces and multi-threading — the Java foundation Selenium scripts run on.

4

Database Testing — MySQL

Phase 4

Database fundamentals, SQL sub-languages, operators, joins, and aggregate functions for verifying data integrity and test data.

5

Web Technologies for Testers

Phase 5

HTML structure & forms and CSS layout fundamentals, so testers can identify, debug and locate UI elements with precision.



02 Program Syllabus

TERM 1 · MANUAL TESTING

FOUNDATIO
NS

Testing Fundamentals & SDLC

- Developers vs. test engineers, defects & testing types
- Waterfall, spiral, V-model & hybrid SDLC models
- White-box testing — path, condition & loop testing
- Black-box testing — functional, integration, regression

Defects, Agile & Documentation

- Defect severity & the defect life cycle
- Scrum, Sprint planning & bug-triaging meetings
- Test plan and test case templates
- Software Testing Life Cycle (STLC)

TOOLS & TECHNOLOGIES

JIRA

Test Plan Templates

Agile/Scrum

TERM 2 · AUTOMATION TESTING (SELENIUM)

AUTOMATIO
N

Selenium WebDriver

- WebDriver architecture & environment setup
- Locators, XPath (relative/absolute) & CSS selectors
- WebElement methods — dropdowns, alerts, frames, windows
- Implicit, explicit & fluent waits

Maven & TestNG

- Maven setup and dependency management
- Writing test cases using TestNG
- Assertions, grouping & data-driven testing
- TestNG listeners and test suites

TOOLS & TECHNOLOGIES

Selenium WebDriver

Maven

TestNG

Eclipse IDE

TERM 3 · CORE JAVA FOR TESTERS

PROGRAMM
ING

Java Essentials

- Java basics & Object-Oriented Programming
- Inheritance and method overriding
- Exception handling — try/catch/finally
- Interfaces and multi-threaded programming

TOOLS & TECHNOLOGIES

Java

Eclipse IDE



TERM 4 · MYSQL & WEB TECHNOLOGIES

DATABASE
+ WEB

MySQL for QA

- Database fundamentals & SQL sub-languages
- Comparison operators and refining selections
- Aggregate functions and SQL files
- MySQL joins — inner, left & right outer

HTML & CSS for Testers

- Page structure, formatting & forms
- Form controls, attributes & elements
- CSS core syntax & selectors
- Layouts — box model, flexbox & grid

TOOLS & TECHNOLOGIES

MySQL Workbench

HTML5

CSS3



03 Real-Time Projects & Case Studies

Instagram Database Clone — Project Test Plan

Write a complete test plan and test cases covering Instagram's core data flows and edge cases.

Hands-on Build

Favorite Blog — Functional Testing Project

Apply black-box functional testing techniques across a blogging application's create/edit/publish flows.

Hands-on Build

Resume Builder (HTML) — UI Testing Project

Validate layout, cross-browser rendering and form behaviour on an HTML-based resume builder.

Hands-on Build

Registration & Login — Test Automation Project

Automate sign-up and login scenarios end-to-end with Selenium WebDriver and TestNG assertions.

Hands-on Build

Paytm Clone Page — Regression Suite Project

Build a regression test suite covering a payments-style interface's critical user paths.

Hands-on Build

Portfolio Page — Cross-Browser QA Capstone

Plan, execute and document a full QA cycle — test plan, manual test cases, and an automated Selenium regression suite.

Capstone Project